

A Foundation-Model T-cell Dysfunction Screen: Translational Candidates, Then a Full Audit

Four checkpoint/persistence hits, tissue-validated -- and what a full concordance, ambient, and donor-level audit does and does not change about them.

Kaisar Dauyey · Shinji Nakaoka

Laboratory of Mathematical Biology, Faculty of Advanced Life Science, Hokkaido University

Contact: k.dauyey.bio.nu@gmail.com

Talk length: within 30 minutes; the time slot itself is not fixed. The main talk (slides 1-21) is budgeted at 23:45 of speaking, leaving about 5 minutes for questions; per-slide budgets are in the deck's speaker notes. The abstract has no stated word limit.

Written report of the IMRaD oral presentation built 2026-09-25 23:57 JST (14:57Z). Every sentence and number below is taken from the deck or its abstract; sources are listed at the end.

Contents

Right-click and choose Update Field to build the table of contents.

Abstract

Background: Foundation models fine-tuned on single-cell transcriptomes can be probed for candidate disease-state drivers by in-silico perturbation. A classifier trained on real tissue, however, also learns whatever else correlates with its labels (ambient RNA, detection artifacts, cohort composition), and a screen alone cannot say which of these it found.

Methods: A donor-disjoint SCLC/LUAD/normal T-cell classifier (46,140 cells, 42 donors; 91.9% accuracy, macro F1 0.903, a third of which comes from one held-out normal-class donor with 566 cells) supported an in-silico deletion and overexpression screen, scored by significance and by bidirectional concordance between the two operations. A screen-level ambient-risk audit and a donor-weighted reanalysis followed the screen.

Results: Four checkpoint and persistence candidates from the original work (TIGIT, TIM-3, CTLA-4, IL7R) replicate by deletion ($FDR < 0.05$, same deletion sign in all three SCLC donors) and validate in independent spatial tissue data (antigen-presentation programme, $\rho = 0.361$, 7.95 sigma above the null). The concordance half of their original call came from a panel runner that overexpressed into all 2,424 SCLC cells but deleted only in detected cells; it has not been rerun on paired cell sets. The original work left a checkpoint-axis ordering open. At donor level, SCLC and LUAD do not differ significantly on that axis ($p = .216$, two-sided Monte Carlo, 100,000 replicates, 19 v 22 donors) once one donor's 74.9% cell share is accounted for. In a later, expanded screen the two most significant hits by FDR (S100A8, S100A9; $FDR \sim 1e-224$) fail bidirectional concordance in all six comparisons tested; screen-wide, half of the top 120 candidates by effect are ambient-flagged, a curated ambient anchor, or both. One candidate, NBEAL1, survives every check applied. Model precision (bf16 vs fp32) replicates cleanly ($\rho > 0.998$) and explains none of these results. The pre-registered 104M sign-agreement gate still reads FAIL, final, because a bitwise-deterministic fp32-vs-fp32 rerun set its noise floor to zero, which is not a usable sign threshold. That FAIL is a property of the floor, not

evidence that bf16 disagrees with fp32, and the domain reviewer ruled to leave the frozen gate unamended.

Discussion: A perturbation screen can support real translational candidates and still require gene-by-gene auditing for confounds. Significance and effect size alone are not sufficient evidence of a cell-intrinsic driver, and an ordering claim built on a cell-weighted cross-cohort comparison needs a donor-level check before it is reported either way.

Introduction

T cells in lung cancer: why they matter, and what was already known

Added 2026-09-28 for a general genetics audience; not in the 25 Sep deck.

The disease

Lung cancer comes in distinct types. **Small cell lung cancer (SCLC)** is the most aggressive thoracic malignancy; **lung adenocarcinoma (LUAD)** is a non-small-cell type.

SCLC itself splits into subtypes defined by master transcription factors (ASCL1, NEUROD1, POU2F3) (Rudin 2019).

The immune angle

T cells can kill tumour cells. Under chronic stimulation they can become dysfunctional ("exhausted"), marked by inhibitory receptors such as PD-1, TIM-3 and CTLA-4 (Wherry 2015).

Immune checkpoint inhibitors release these brakes, but add only a modest survival benefit in extensive-stage SCLC (Horn 2018), despite its high tumour mutational burden.

The open question

Whether SCLC T cells are dysfunctional, and through which genes, was poorly characterised at single-cell resolution when this work began.

SCLC subtypes differ in immune activation; an inflamed subtype benefits most from immunotherapy (Gay 2021). This talk asks what a model trained on T cells themselves can say.

In plain words: we study the immune cells inside lung tumours, because drugs that re-activate them help SCLC patients less than hoped, and nobody knows exactly why.

Sources: SCLC 'most aggressive thoracic malignancy', 'responds poorly to immune checkpoint blockade despite a high tumour mutational burden', 'transcriptional basis poorly characterised at single-cell resolution': JSDP poster, section A 'Why this matters' - data/sources/poster_final.pdf (our 24JSDP poster, P25; geneformer-lung-tcell poster/poster_final.pdf, untracked; sha256 94638dba...) | SCLC transcription-factor subtypes: Rudin CM, Poirier JT, Byers LA, et al. Molecular subtypes of small cell lung cancer: a synthesis of human and mouse model data. Nat Rev Cancer 2019. doi:10.1038/s41568-019-0133-9 | T-cell exhaustion and inhibitory receptors: Wherry EJ, Kurachi M. Molecular and cellular insights into T cell exhaustion. Nat Rev Immunol 2015. doi:10.1038/nri3862 | Modest benefit of adding atezolizumab to chemotherapy in extensive-stage SCLC: Horn L, Mansfield AS, Szczyrna A, et al. First-line atezolizumab plus chemotherapy in extensive-stage small-cell lung cancer. N Engl J Med 2018. doi:10.1056/NEJMoa1809064 | Inflamed SCLC subtype and immunotherapy benefit: Gay CM, Stewart CA, Park EM, et al. Patterns of transcription factor programs and immune pathway activation define four major subtypes of SCLC with distinct therapeutic vulnerabilities. Cancer Cell 2021. doi:10.1016/j.ccell.2020.12.014

Our 24JSDP poster (P25) and what the audit changed

Added 2026-09-28 for a general genetics audience; not in the 25 Sep deck.

Our 24JSDP poster (P25) reported a donor-held-out classifier (91.9% accuracy), four replicated edits (TIM-3, TIGIT, CTLA-4, IL7R) and tissue validation in 5 Visium SCLC sections (15,632 spots).

Poster statement	Status after the audit (this talk)
Four edits concordant, FDR < 0.05 in both arms, same sign in all 3 SCLC donors	Deletion replication stands. The concordance half was scored on different cell sets per arm (overexpress 2,424 cells, delete 202-

	1,131) and has not been rerun.
"The model implies Normal < SCLC < LUAD on the checkpoint axis ... an open question"	Closed as null: donor-level SCLC vs LUAD, p = 0.216.
Detection vs effect size, rho = -0.60: sparse genes fake big effects	Unchanged; the original talk's own QC slide, shown here as Results R2.
Tissue: dysfunction tracks T-cell abundance (rho 0.161); talk: antigen presentation rho 0.361	Unchanged; not re-analysed here.

In plain words: the poster's main hits survive; one of its claims (an ordering of the three disease states) did not hold up once each person was counted once.

Sources: Poster title, number, authors, sections and all poster quotes: data/sources/poster_final.pdf (our 24JSDP poster, P25; geneformer-lung-tcell poster/poster_final.pdf, untracked; sha256 94638dba...) | rho 0.361 / 7.95 sigma: data/sources/JSDP_P25_talk.pdf (the talk that accompanied the poster; geneformer-lung-tcell talk/JSDP_P25_talk.pdf, untracked; sha256 3f013a5d...) (slide 9). | Status column restates this deck's own slides: Results R1 (cell-set caveat; targeted_panel_delete_overexpress_merged.csv @ 6882627, runner fix 66d235b) and R3 (t6_permutation_tests.csv @ 9ec518c). | Held-out cell count: poster 9,377 vs confusion matrix 9,376 is a known one-cell truncation with no effect at any reported precision (METHODS.md lines 125-166) - not an audit change. | Per the human's ruling (2026-09-28), the poster is named only as 'our 24JSDP poster (P25)': no venue, date or meeting details.

A perturbation screen can look statistically overwhelming and still be wrong

Foundation models fine-tuned on single-cell transcriptomes can be probed for candidate disease-state drivers: delete or overexpress a gene's simulated expression in a held-out cell, and read the shift in a fine-tuned classifier's predicted state. A dose-responsive driver should move the model toward its target when overexpressed and away from it when deleted.

A classifier trained on real tissue also learns whatever else correlates with its labels -- ambient RNA from other cell types, detection-rate differences, cohort composition, tissue of origin. None of that is cell-intrinsic biology, and a screen does not tell you on its own which one it found.

This talk presents a set of translational candidates identified this way, then applies the audit a screen like this needs: a concordance test, a screen-level ambient-contamination check, and a donor-level reanalysis of a cross-cohort ordering claim.

Four replicated checkpoint/persistence hits, tissue-validated

A donor-disjoint SCLC/LUAD/normal T-cell classifier supported an applied in-silico perturbation screen for candidate CAR-T/ICI-relevant drivers. Four hits replicated across the screen: **TIGIT**, **TIM-3** (**HAVCR2**), **CTLA-4**, **IL7R**.

Independent Visium spatial validation (GSE263196, 5 SCLC specimens, 15,632 in-tissue spots) supported a tissue-validated antigen-presentation programme.

Presented at JSDP, 2026-08-17. No caveat attached here -- none of these four hits is implicated in anything that follows.

b

SCLC T cells → normal T-cell state:
4 knockouts replicate across
all 3 SCLC donors

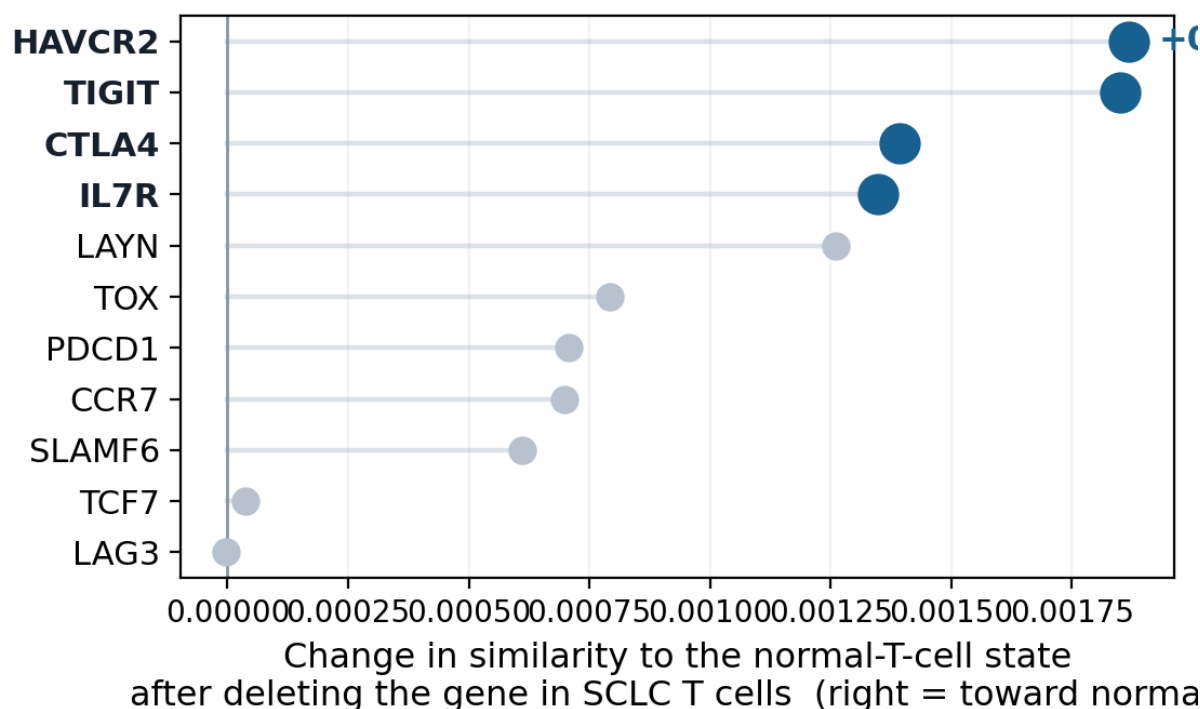


Figure 1. Slide 3 of the 25 Sep deck: Four replicated checkpoint/persistence hits, tissue-validated. Source: talk/JSDP_P25_talk.pptx; poster/poster_final.html. Image: data/assets/screen_b.png.

Source (slide 3 of the 25 Sep deck): talk/JSDP_P25_talk.pptx; poster/poster_final.html

One test was already flagged as pending; one new screen needed the same scrutiny

The poster's own hedge	The talk's own QC slide, "TESTED SINCE THE POSTER"	What this talk adds
"The model implies Normal < SCLC < LUAD on the checkpoint axis. This conflicts with the clinical tumour-level picture of SCLC as cold and ICI-resistant, so it remains an open question rather than a reconciled conclusion. "	"The states are not on a line, so there is no ordering to contradict... Test 1 of 4 — sign directions, not centroid geometry. Pending T2 and T3. " The donor-level test was named as open before this work began.	A later, expanded screen surfaced two new high-FDR candidates needing the same concordance scrutiny the original four received. This talk completes the pending test and applies that scrutiny — it does not reopen a settled claim.

Source (slide 4 of the 25 Sep deck): poster/poster_final.html; talk/JSDP_P25_talk.pdf ("TESTED SINCE THE POSTER", p.7)

Methods

Three tools, one analogy each

Added 2026-09-28 for a general genetics audience; not in the 25 Sep deck.

Single-cell RNA sequencing

Reads which genes are switched on in **each individual cell**, not in a ground-up tissue sample.

Analogy: a list of every fruit in the bowl, instead of the smoothie.

Here: 46,140 T cells from 42 people (HTAN MSK collection, CELLxGENE).

Geneformer, a foundation model

A neural network pre-trained on a large corpus of single-cell profiles (Theodoris 2023). It reads a cell as its genes **ranked by expression**, most-expressed first, and turns it into a list of numbers (an **embedding**) that puts similar cells near each other.

Analogy: a map coordinate for every cell. We fine-tuned it to tell SCLC, LUAD and normal-tissue T cells apart.

In silico perturbation

Delete one gene from a cell's ranked list, or move it to the top ("overexpress"), re-read the cell, and measure how far the model's picture of it moves toward another disease state.

Analogy: a Perturb-seq screen run on the computer. **Nothing is edited in a lab**: it generates hypotheses, it is not a knockout.

In plain words: we taught a model what SCLC, LUAD and normal T cells look like, then asked it "what if this gene were missing, or boosted?" for thousands of real cells.

Sources: 46,140 cells, 42 individuals, HTAN MSK / CELLxGENE dataset 6fde3ad9:

`data/git/dd7366c/sclc_validation/perturbation_workflow/METHODS.md` (geneformer-lung-tcell @ dd7366c) (lines 5-10) and the poster's cohort panel. | Geneformer: Theodoris CV, Xiao L, Chopra A, et al. Transfer learning enables predictions in network biology. Nature 2023. doi:10.1038/s41586-023-06139-9 | 'Reads a cell as an ordered list of its genes, most-expressed first'; 'nothing is edited in a laboratory'; 'a hypothesis generator, not a knockout experiment'; 'a Perturb-seq screen run in silico': JSDP poster Fig. 1 and Fig. 5 captions - `data/sources/poster_final.pdf` (our 24JSDP poster, P25; geneformer-lung-tcell poster/poster_final.pdf, untracked; sha256 94638dba...) | Delete = token removed from the rank-value encoding; overexpress = moved to the front: `data/git/dd7366c/sclc_validation/perturbation_workflow/METHODS.md` (geneformer-lung-tcell @ dd7366c) (lines 180-187).

46,140 T cells, 42 donors, one fine-tuned classifier

Held-out test performance

91.9% accuracy, macro F1 0.903, donor-disjoint split. Geneformer V2 104M fine-tune.

Normal-class test performance rests on 1 held-out donor (566 cells)

A third of this macro-F1. Already flagged in the repo's own methods (perturbation_workflow/METHODS.md, lines 57-67 the split table, line 112: "Normal-class metrics are flagged explicitly given 1 test donor").

19 SCLC + 22 LUAD + 4 normal = 45, but 42 unique donors: three donor_ids (RU675, RU682, RU684) each carry two biospecimens from the same clinical patient, one LUAD tumour and one normal-tissue sample (METHODS.md, lines 33-39).

In plain words: shown T cells from 8 people it had never seen, the model named the right disease state for about 92 of every 100 cells; the normal-tissue part of that score rests on one person.

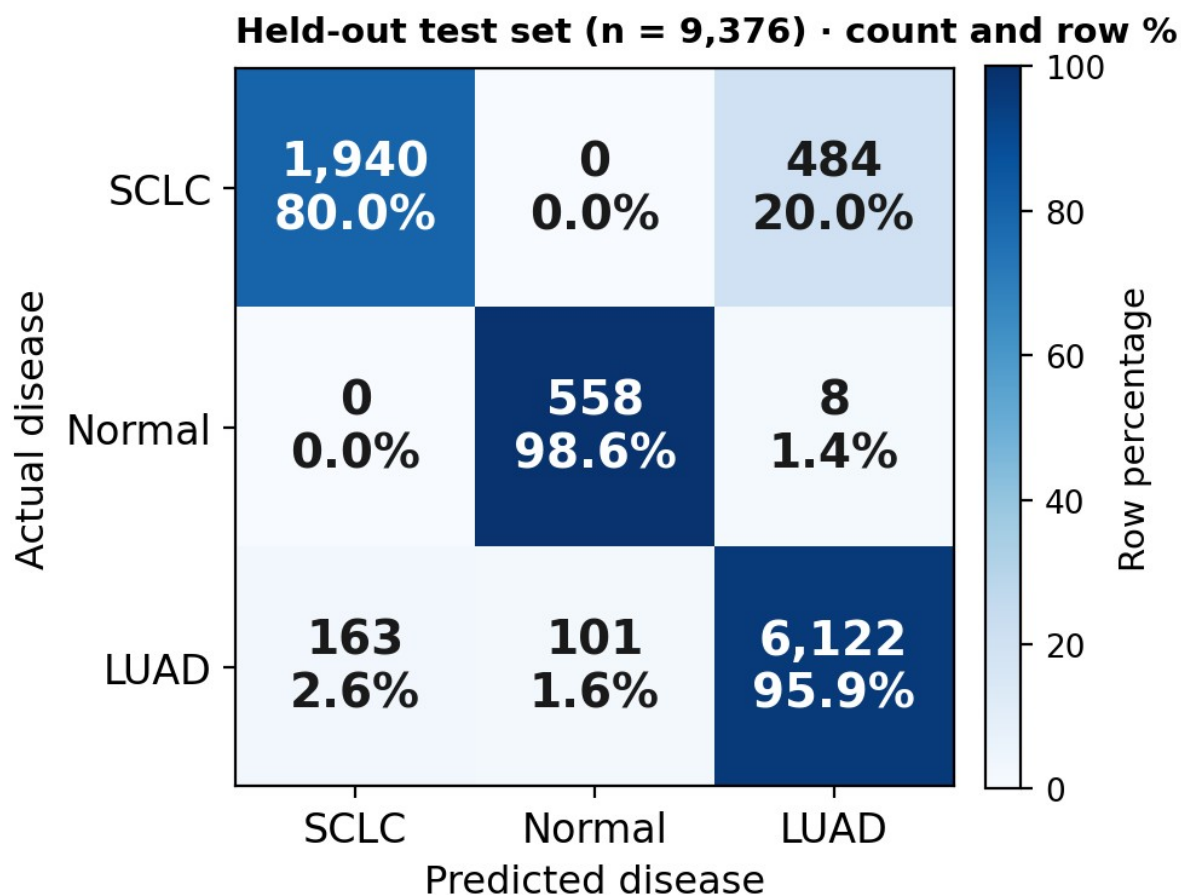


Figure 2. Slide 5 of the 25 Sep deck: 46,140 T cells, 42 donors, one fine-tuned classifier. Source: README.md; perturbation_workflow/METHODS.md. Image: data/assets/confusion.png.

Source (slide 5 of the 25 Sep deck): README.md; perturbation_workflow/METHODS.md

Delete or overexpress, in held-out cells, across six comparisons

Each gene is deleted (its token removed from a cell's rank-ordered input) or overexpressed (promoted to the front of the rank) in held-out cells, and the shift in the classifier's predicted disease-state probability is read for both operations.

Six source→goal disease-state comparisons: SCLC→LUAD, SCLC→normal, LUAD→SCLC, LUAD→normal, normal→SCLC, normal→LUAD.

A dose-responsive driver should move the model toward its target under overexpression and away from it under deletion — the two operations are read together, never one alone.

Source (slide 6 of the 25 Sep deck): README.md

A real driver must move opposite ways under deletion and overexpression

The concordant criterion, used throughout this repo's perturbation work:

```
delete_fdr < 0.05 AND overexpress_fdr < 0.05 AND delete_shift ×
overexpress_shift < 0 AND delete_n ≥ 25
```

Significance in one arm is not enough. A gene must clear both arms and reverse sign between them — the signature a dose-responsive driver actually produces. A gene can be significant in both arms and still fail this test if deletion and overexpression push the classifier the **same** direction — the model reacting to the gene's presence at all, not to a dose-responsive relationship with the labelled state.

In plain words: a gene counts only if removing it and boosting it push the cell in opposite directions, each beyond chance; $FDR < 0.05$ means at most about 5% of such calls are expected to be false.

Source (slide 7 of the 25 Sep deck): `primary_test_perturbation/scripts/build_delete_overexpress_shift_report.py` @ b99365b

Two further checks, applied after the screen, not instead of it

Ambient-risk audit	Donor-weighted reanalysis	Why both, not either
Every top-effect gene is checked against a curated ambient-contamination panel and a curated T-cell-intrinsic anchor panel. The flag threshold is the 25th percentile of the full ambient-anchor population's own risk scores — a within-panel, not an absolute, cutoff.	Any cross-cohort comparison is recomputed donor-level (mean within donor first, then an unweighted mean of donor means) alongside the original cell-weighted statistic, to separate a real group difference from one donor's cell count dominating the mean.	A significance test alone cannot distinguish a cell-intrinsic effect from an ambient or cell-count artifact. Both audits are applied to every candidate discussed in Results, not selectively to ones that already look questionable.

Source (slide 8 of the 25 Sep deck): `ambient_risk_diagnostic.py` line 198 @ b99365b; `immune_axis_test/donor_robustness.py` @ 9ec518c

Results

Four hits replicate and validate independently in tissue

TIGIT, TIM-3 (HAVCR2), CTLA-4, IL7R

Four replicated checkpoint/persistence edits from the applied screen (`screen_b.png`, Introduction). Replication here means: deletion $FDR < 0.05$ and the same deletion sign in all 3 SCLC donors.

GSE263196 Visium, 5 SCLC specimens, 15,632 in-tissue spots: antigen-presentation programme $\rho = 0.361$, 7.95 sigma above the null.

The concordance half of the original call was made by the pre-2026-09-22 panel runner, which overexpressed into all 2,424 SCLC cells while deleting only in the 202–1,131 cells expressing each gene; it has not been rerun with paired cell sets. Unaffected by S100, ambient or T6 below; the spatial number is untouched. A qualifier, not a retraction.

In plain words: in real tumour slices, spots with more antigen-presentation activity also held more T cells ($\rho 0.361$, where 0 is no relation and 1 is perfect), far beyond chance (7.95 sigma).

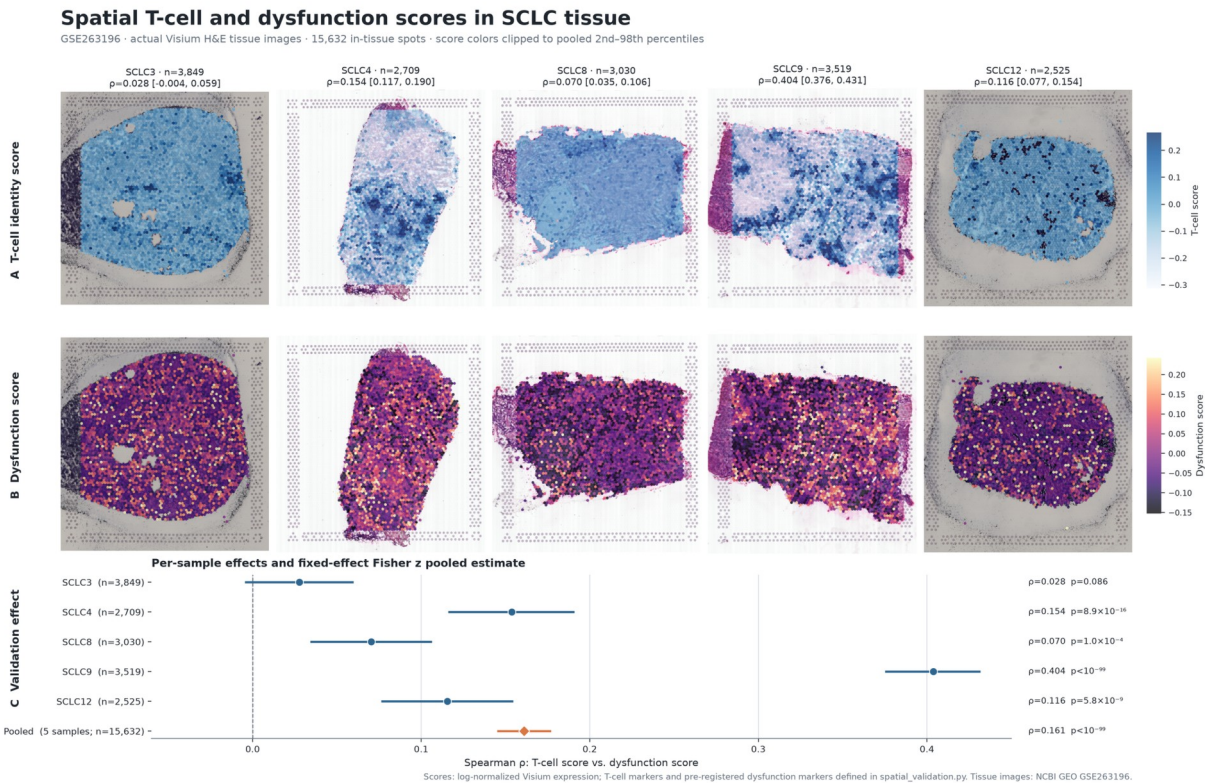


Figure 3. Slide 9 of the 25 Sep deck: Four hits replicate and validate independently in tissue. Source: talk/JSDP_P25_talk.pptx/pdf (rho, sigma: talk only); targeted_panel_delete_overexpress_merged.csv @ 6882627; runner fix 66d235b. Image: data/assets/spatial.png.

Source (slide 9 of the 25 Sep deck): talk/JSDP_P25_talk.pptx/pdf (rho, sigma: talk only); targeted_panel_delete_overexpress_merged.csv @ 6882627; runner fix 66d235b

Sparse genes fake big effects, and the poster's own ordering fails its own test

Detection vs. effect size: Spearman rho = -0.60 ($p = 2.7e-5$, $n = 42$) across the full 50-gene panel.

“TESTED SINCE THE POSTER” (talk, slide 7): “the states are not on a line, so there is no ordering to contradict... Working reading: a triangle — SCLC $\leq$ Normal < LUAD... Test 1 of 4 — sign directions, not centroid geometry. **Pending T2 and T3.**”

This is the original team's own self-correction, dated before this work began — not a finding of ours.

In plain words: the rarer a gene is in the data, the bigger its apparent effect tends to be (rho -0.60), so the size of an effect alone is not evidence.

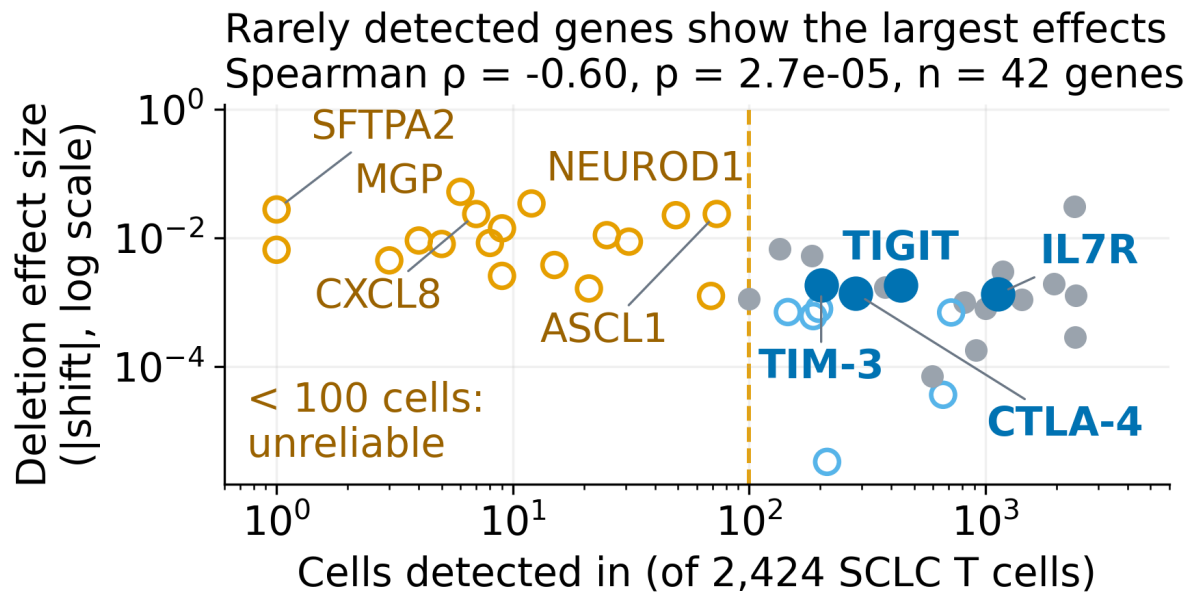


Figure 4. Slide 10 of the 25 Sep deck: Sparse genes fake big effects, and the poster's own ordering fails its own test. Source: talk/JSDP_P25_talk.pdf, slide 7. Image: data/assets/detection.png (readable redraw of 2026-09-28, same data; original: data/assets/detection_orig.png).

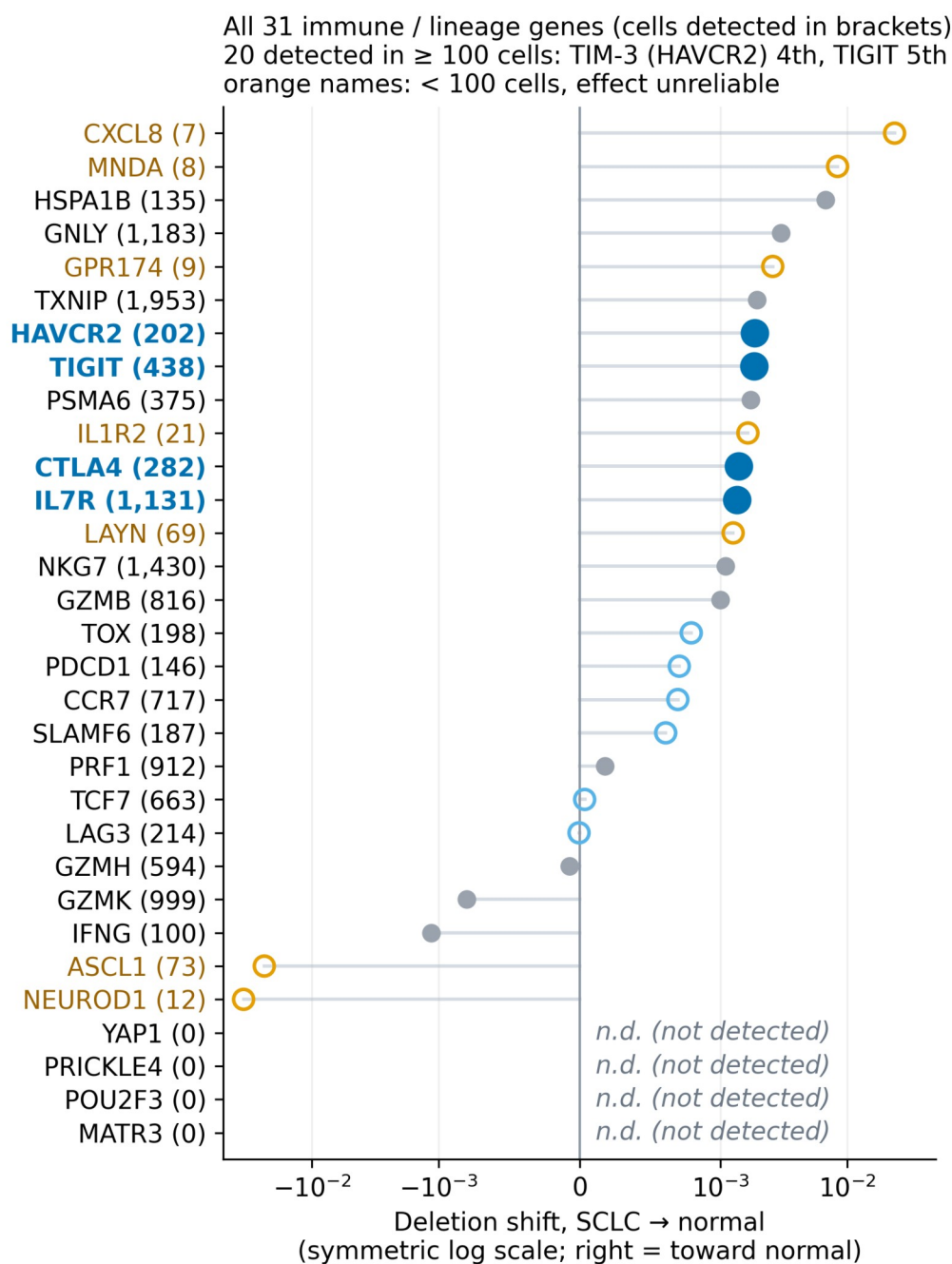


Figure 4 (continued), panel b: all 31 immune / lineage genes ranked by deletion shift, SCLC → normal. Panel b of the same original figure, drawn separately so its gene names stay readable. Image: data/assets/detection_rank_report.png.

Source (slide 10 of the 25 Sep deck): [talk/JSDP_P25_talk.pdf](#), slide 7

The open question is now closed as null

Donor-level: no reversal

Complete population: SCLC 0.301 > LUAD 0.244. Test-only: SCLC 0.321 > LUAD 0.267. Two-sided Monte Carlo test, 100,000 replicates: $p = 0.216$, 19 vs. 22 donors. **Not significant.**

PleuralEffusion: one genuine donor

Carries 74.9% of test-only SCLC cells. HTAN_Participant_ID confirms exactly one participant (HTA8_2001), not several merged (METHODS.md, lines 29-32).

A separate, **exact-enumeration** test on the 3-vs-4-donor test-only slice: only 35 label assignments exist; the smallest attainable two-sided p is 0.0571 (2/35) — that design cannot reach significance regardless of the true effect.

In plain words: counting each person once instead of each cell, SCLC and LUAD T cells do not differ significantly on this axis ($p = 0.216$); one donor had supplied most of the test-only SCLC cells.

The SCLC-vs-LUAD "disagreement" is a weighting artifact

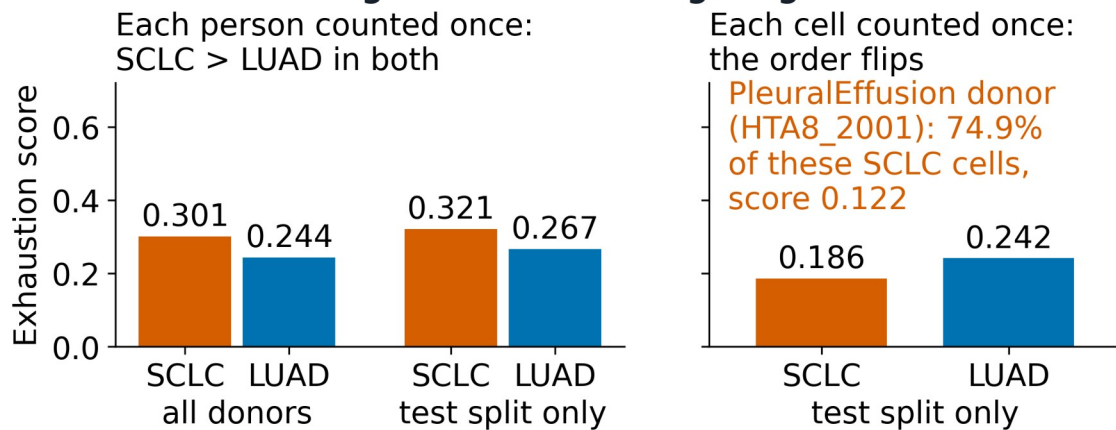


Figure 5. Slide 11 of the 25 Sep deck: The open question is now closed as null. Source: [t6_weighting_reconciliation.csv](#), [t6_permutation_tests.csv](#) @ 9ec518c. Image: [data/assets/fig_t6_reversal.png](#) (readable redraw of 2026-09-28, same data; original: [data/assets/fig_t6_reversal_orig.png](#)).

Source (slide 11 of the 25 Sep deck): [t6_weighting_reconciliation.csv](#), [t6_permutation_tests.csv](#) @ 9ec518c

S100A8 and S100A9: an overwhelming FDR, and no concordance at all

9 of 12 rows same-sign; 0 of 6 comparisons concordant

Example: S100A8, LUAD→SCLC deletion — FDR = $1.45e-224$, N = 713 detections, shift = +0.0255. **The FDR is a certainty claim, not a size claim** — it says nothing about direction. First appearance of these two genes in this deck: caveated from the start, not a later correction.

What the LUAD-normal and SCLC-normal rows can and cannot lean on

Of 12 rows, 8 touch the normal class (4 comparisons). For $\text{luad} \leftrightarrow \text{normal}$, 3 of 4 normal donors are the *same patients* as 3 LUAD donors — real protection against a separate-cohort confound, though only 3 of 4. **No such pairing exists for $\text{sclc} \leftrightarrow \text{normal}$.**

In plain words: an FDR of $1.45e-224$ says the shift is almost certainly not chance, not that it is large or meaningful; S100A8/S100A9 still fail the both-directions test in all 6 comparisons.

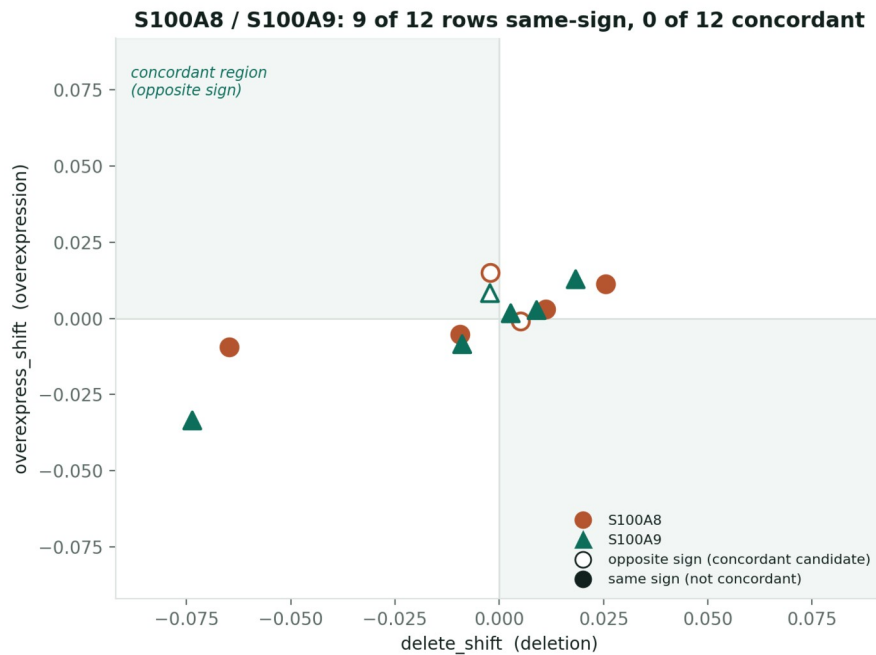


Figure 6. Slide 12 of the 25 Sep deck: S100A8 and S100A9: an overwhelming FDR, and no concordance at all.
Source: [isp_plausibility_top_candidates.csv @ b99365b](#). Image: [data/assets/fig_concordance.png](#).

Source (slide 12 of the 25 Sep deck): [isp_plausibility_top_candidates.csv @ b99365b](#)

Half the top 120 is ambient-flagged or a curated ambient anchor

60/120

rows are ambient-flagged, a curated ambient anchor, or both: 31 flagged-only, 5 anchor-only, 24 both — a real partition of the 120, not two overlapping counts added together

5/120

rows are anchors below the flag threshold — mechanical, not a miss: the threshold is the anchors' own 25th percentile of risk, so some anchors falling under it is built in.

0/36

of the 36 curated T-cell-intrinsic anchor genes (a different denominator: the full curated list, not a row count) appear anywhere in the 120.

In plain words: half of the 120 strongest hits carry a flag for possible contamination by RNA from other cells, or are known contaminant marker genes, so they need caution before being called T-cell biology.

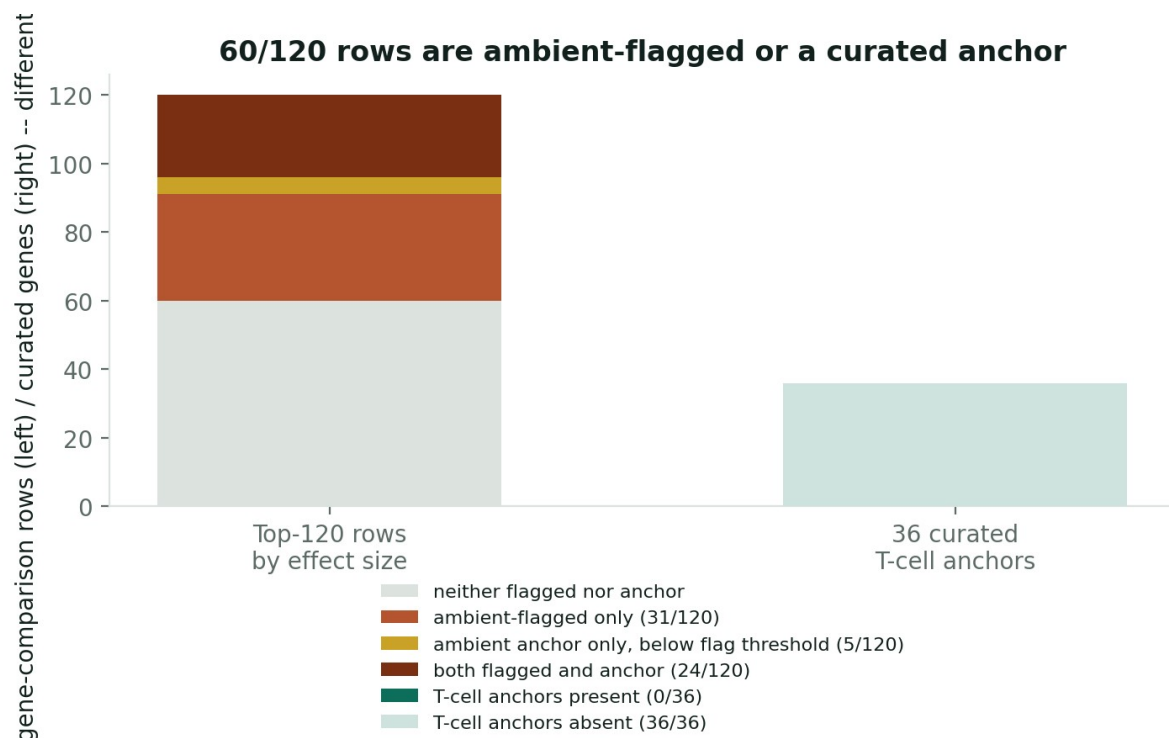


Figure 7. Slide 13 of the 25 Sep deck: Half the top 120 is ambient-flagged or a curated ambient anchor.
Source: `top_n_by_effect` rows, `ambient_risk_manifest.json` @ b99365b. Image: `data/assets/fig_ambient_breakdown.png`.

Source (slide 13 of the 25 Sep deck): `top_n_by_effect` rows, `ambient_risk_manifest.json` @ b99365b

One candidate survives every check

NBEAL1

5 of 5 comparisons concordant — every one it appears in — detected in 75.0% of cells.

Passes the same concordance test that eliminates S100A8/S100A9, at a detection rate well above the ambient range flagged for those two. Does not make it a validated driver on its own — makes it the one candidate this screen has not been able to eliminate.

Not everything restored from the original screen fails, and the one thing that doesn't fail is more informative for having survived a real test.

In plain words: NBEAL1 passed the both-directions test in all 5 comparisons it appears in and is seen in 3 of every 4 cells, so it is not a rare-gene artefact; it is a candidate, not a proven driver.

Source (slide 14 of the 25 Sep deck): `isp_plausibility_top_candidates.csv` @ b99365b

S100 genes did not beat their matched control genes

Added 2026-09-28 for a general genetics audience; not in the 25 Sep deck.

Question and design

Do simulated deletion and overexpression of S100 genes push LUAD T cells toward normal or SCLC **more than matched control genes** do, and more so for the 4 S100 genes at high ambient-RNA risk (S100A2, S100B, S100A13, S100PBP) than for 4 at low risk (S100A4, S100A6, S100A10, S100A11)?

Each gene vs its own 20 detection-matched controls; the two groups of 4 compared with an exact Mann-Whitney test; all registered in advance.

Result: negative in both comparisons

LUAD → normal: U = 1.5, p = 6/70 (0.086).
LUAD → SCLC: U = 5.5, p = 34/70 (0.49).

Stability gate not positive (0/160 leave-one-control-out). No-op check passed. S100P and S100A16 not estimable (40 and 48 cells, below the 50-cell gate). S100A8/S100A9 excluded by design (classifier anchors).

How to read it

The test was small: with 4 v 4 the smallest possible p is 2/70 (0.029), reached only by complete separation.

The ambient-high genes did not beat matched controls more than the ambient-low genes. No claim is made from the ambient-low genes' high scores.

This **neither confirms nor contradicts** the S100A8/S100A9 slides.

In plain words: in a small, pre-registered test, S100 genes flagged for possible contamination did not stand out against look-alike control genes.

Sources: NEW since the 25 Sep deck; pre-registered; run 2026-09-28. | Status, U, p, stability gate, no-op PASS, S100P/S100A16 not_estimable, anchors S100A8/A9: data/git/53adee2/.../results_s100_luad_20260928/RESULT.md (geneformer-lung-tcell @ 53adee2; merged to main in PR #37, 4455914); numbers read at build time from data/git/53adee2/.../results_s100_luad_20260928/primary_results.json (@ 53adee2) | Gene groups and ambient risk: data/git/53adee2/.../s100_isp/analyse_s100_luad_20260928.py (@ 53adee2): gene groups, Ensembl IDs, ambient-risk values | Wording rules (never 'no effect'; state the 2/70 floor; no claim from clean genes' high Q): the lab's ISP outcome standard (ISP-STD-1 v1.1, the floor's wording rules; not in this package) | Mann-Whitney test: Mann HB, Whitney DR. On a test of whether one of two random variables is stochastically larger than the other. Ann Math Stat 1947. doi:10.1214/aoms/1177730491

Discussion

What this establishes, and what it rules out

Establishes	Resolves	Rules out
Four checkpoint/persistence hits (TIGIT, TIM-3, CTLA-4, IL7R) replicate and validate independently in spatial tissue data. A concordance test and a screen-level ambient audit are necessary companions to significance for any further perturbation-based driver claim. One further candidate (NBEAL1) clears every check applied.	The checkpoint-axis ordering the original team flagged as open ("pending T2 and T3") is closed: donor-level SCLC-vs-LUAD shows no significant difference in either direction, once a single donor's cell-count share is accounted for.	S100A8/S100A9 as T-cell-intrinsic dysfunction drivers, at high confidence, independent of model precision. Model precision (bf16) as an explanation for any finding in this deck.

The S100 ISP result in context (added 2026-09-28)

The pre-registered S100 in silico perturbation test (Results) was negative in both LUAD comparisons: the four S100 genes at high ambient-RNA risk did not beat their detection-matched controls more than the four at low risk. The test was small (4 v 4; the smallest attainable p is 2/70), so it could only have passed on a complete separation of the two groups, and no claim is drawn from the low-risk genes' high scores. It neither confirms nor contradicts the S100A8/S100A9 findings above: those two genes are classifier anchors and were deliberately excluded from the test.

Sources: data/git/53adee2/.../results_s100_luad_20260928/RESULT.md (geneformer-lung-tcell @ 53adee2; merged to main in PR #37, 4455914); the lab's ISP outcome standard (ISP-STD-1 v1.1, the floor's wording rules; not in this package)

Limitations

What this study does not establish

Detection-confound axis is an imperfect proxy	Single cohort; curated lists	Thin normal class, both counts	Tissue-of-origin skew, per contrast
Ambient risk is scored partly against the classifier being audited — evidence of resemblance to known contaminants, not a direct contamination measurement.	One HTAN/CELLxGENE cohort; the ambient-anchor and T-cell-anchor gene lists are curated, not ground truth. A different cohort or anchor set could shift specific gene calls.	1 held-out test donor AND 4 cohort donors overall — already a named limitation in the repo's own methods (METHODS.md, lines 24-25), carried forward here, not newly flagged.	SCLC: 7/19 (37%) lung, 12/19 metastatic. LUAD: 16/22 (73%) lung. Normal: 4/4 (100%) lung. Skews sclc_to_normal/normal_to_sclc, not luad_to_normal/normal_to_luad (which benefits from the 3-of-4 shared-patient pairing above).

Open questions: the original talk's CRISPR TIGIT/HAVCR2 knockout proposal; does NBEAL1 replicate independently; the talk's own still-open question — “if SCLC T cells are not exhausted, what are they, and why does ICI still fail?” A clean negative, honestly obtained, is a contribution.

Source (slide 18 of the 25 Sep deck): *perturbation_workflow/METHODS.md* lines 24-25; *talk/JSDP_P25_talk.pdf*, slide 11

Glossary

Terms used in this talk

Added 2026-09-28 for a general genetics audience; not in the 25 Sep deck.

Term	Meaning here
Single-cell RNA-seq	Measures which genes are active in each individual cell.
T-cell dysfunction / exhaustion	A worn-out T-cell state after chronic stimulation, with inhibitory receptors (PD-1, TIM-3, CTLA-4) (Wherry 2015).
Immune checkpoint	An inhibitory receptor that brakes T cells; checkpoint-inhibitor drugs block it.
Foundation model (Geneformer)	A model pre-trained on many cells, then fine-tuned for one task (Theodoris 2023).
Embedding	The list of numbers the model uses to place a cell; similar cells sit close together.
In silico perturbation	Deleting or boosting a gene in the model's input and measuring the shift.
Concordant	Deletion and overexpression move the cell in opposite directions, both significant.
FDR	False discovery rate: the expected share of false calls among all calls (Benjamini 1995).
Donor-level analysis	Each person counted once, so one person's many cells cannot dominate.
Ambient RNA	Free-floating RNA from other cells that contaminates a cell's reads (Young 2020).
Matched controls	Non-S100 genes chosen to be detected about as often as the gene tested.
Mann-Whitney U	A rank test of whether one group tends to score higher than another (Mann 1947).
Pre-registration	Fixing the test and its pass/fail rule in writing before seeing the result.
bf16	A lower-precision number format; used to check that rounding does not drive results.

Sources: Glossary definitions are this talk's working definitions. Literature: Wherry EJ, Kurachi M. Molecular and cellular insights into T cell exhaustion. *Nat Rev Immunol* 2015. doi:10.1038/nri3862; Theodoris CV, Xiao L, Chopra A, et al. Transfer learning enables predictions in network biology. *Nature* 2023. doi:10.1038/s41586-023-06139-9; Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J R Stat Soc B* 1995. doi:10.1111/j.2517-6161.1995.tb02031.x; Young MD, Behjati S. SoupX removes ambient RNA contamination from droplet-based single-cell RNA sequencing data. *GigaScience* 2020. doi:10.1093/gigascience/giaa151; Mann HB, Whitney DR. On a test of whether one of two random variables is stochastically larger than the other. *Ann Math Stat* 1947. doi:10.1214/aoms/1177730491

Appendix

Deep-dive material kept for questions; in the talk these are backup slides.

No: precision replicates cleanly; model size is the axis that moves things

Same model, fp32 vs. bf16 (half precision)

104M panel (300 rows): $\rho = 0.9998$, sign agreement 298/300. 316M canary (60 rows): $\rho = 0.9987$, sign agreement 60/60.

The pre-registered 104M sign-agreement gate is still recorded **FAIL** — a bitwise-deterministic fp32-vs-fp32 rerun made the noise floor zero, not a usable sign threshold; the domain reviewer's ruling was that the frozen gate stands unamended, not that bf16 disagrees with fp32. Both are true at once.

Same model, changed precision vs. changed size

Precision (104M fp32→bf16): $\rho = 0.9998$. Size (104M→316M, both bf16): $\rho = 0.489$. Using $(1-\rho)$ as the divergence measure, size moves results $\sim 2,500\times$ (~ 3.4 orders of magnitude) more than precision. None of this deck's findings are precision artifacts.

In plain words: rerunning the model at lower numerical precision gave essentially the same answers ($\rho = 0.9998$), so rounding does not explain any result in this talk.

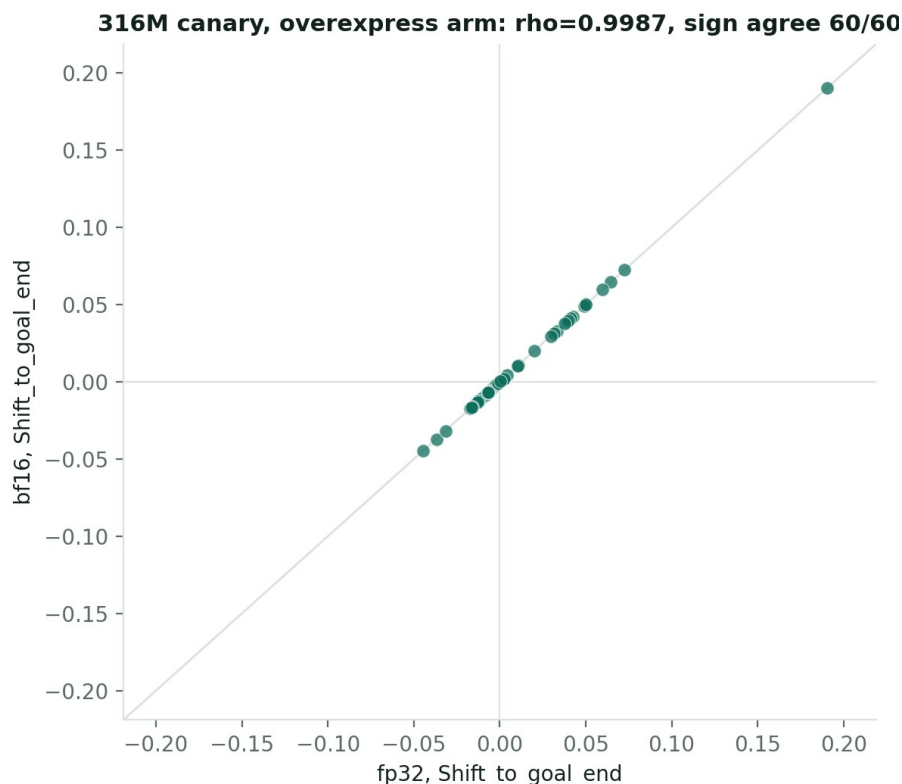


Figure 8. Slide 15 of the 25 Sep deck: No: precision replicates cleanly; model size is the axis that moves things. Source: `committed_run_stats/{bf16,316m_bf16,...}` @ origin/main; RESULTS_BF16.md @ afa0564. Image: `data/assets/fig_bf16_canary.png`.

Source (slide 15 of the 25 Sep deck): `committed_run_stats/{bf16,316m_bf16,...}` @ origin/main; RESULTS_BF16.md @ afa0564

The four original hits sit in one connected interaction neighbourhood

From prior literature, not inferred from these cells

TIGIT: CAR-T support. TIM-3: solid-tumour knockout precedent. CTLA-4: real signal, less CAR-T-specific. IL7R: persistence, not exhaustion. 54 STRING edges among 16 context genes; TOX and LAYN have no edge above threshold; PD-L1 is absent from the T-cell atlas.

No caveat — these four hits are unaffected by anything in this deck's audit.

In plain words: the links between these genes come from STRING, a public database of known and predicted protein interactions (Szkarczyk 2023), i.e. from prior literature, not from these cells.

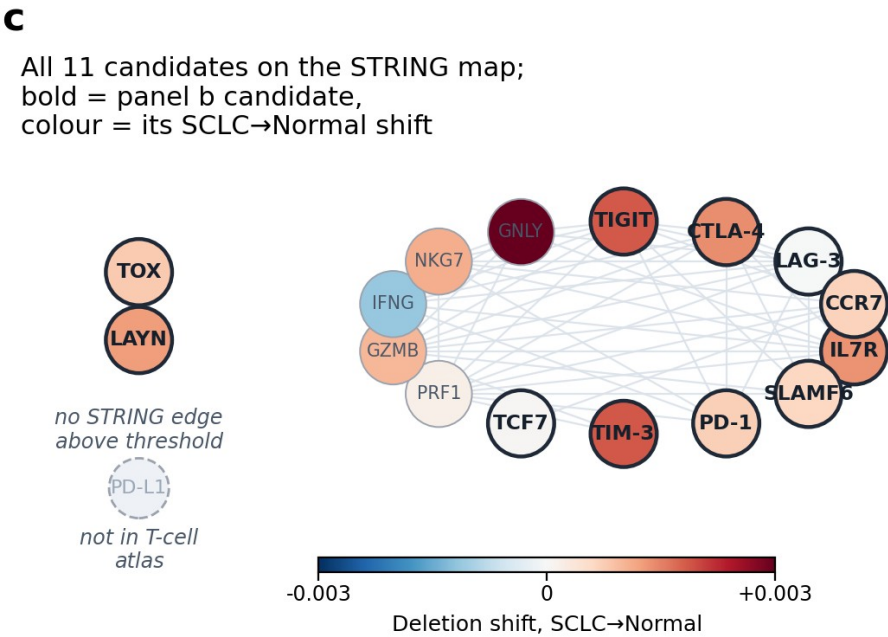


Figure 9. Slide 16 of the 25 Sep deck: The four original hits sit in one connected interaction neighbourhood. Source: talk/JSDP_P25_talk.pdf, slide 10 (STRING literature overlay). Image: data/assets/network_c.png.

Source (slide 16 of the 25 Sep deck): talk/JSDP_P25_talk.pdf, slide 10 (STRING literature overlay)

Each S100 gene's percentile against its own 20 matched controls (Q)

Added 2026-09-28 for a general genetics audience; not in the 25 Sep deck.

Gene	Ensembl ID	Group	Ambient risk	Q, LUAD→normal	Q, LUAD→SCLC
S100A2	ENSG00000196754	ambient-high	0.955	88.1	97.6
S100B	ENSG00000160307	ambient-high	0.859	21.4	16.7
S100A13	ENSG00000189171	ambient-high	0.796	26.2	59.5
S100PBP	ENSG00000116497	ambient-high	0.744	54.8	78.6
S100A4	ENSG00000196154	ambient-low	1.1e-06	88.1	97.6
S100A6	ENSG00000197956	ambient-low	3.4e-05	97.6	88.1
S100A10	ENSG00000197747	ambient-low	0.000312	92.9	88.1
S100A11	ENSG00000163191	ambient-low	0.000144	64.3	50.0

Group test (exact two-sided Mann-Whitney, 4 v 4): LUAD→normal U = 1.5, p = 6/70; LUAD→SCLC U = 5.5, p = 34/70. Negative in both. The ordering seen (ambient-low above) is descriptive only; the ambient-low genes' high Q was not tested and supports no claim.

Sources: Q values and group test read at build time from `data/git/53adee2/.../results_s100_luad_20260928/primary_results.json` (@ 53adee2) | Ensembl IDs, groups and ambient-risk values: `data/git/53adee2/.../s100_isp/analyse_s100_luad_20260928.py` (@ 53adee2): gene groups, Ensembl IDs, ambient-risk values | Q is the gene's midrank percentile among its 20 matched controls; 8 of 8 genes have 20 valid controls in both contrasts.

References and sources

Literature cited in the added sections (each DOI checked on Crossref, 2026-09-28):

1. Theodoris CV, Xiao L, Chopra A, et al. Transfer learning enables predictions in network biology. *Nature* 2023. doi:10.1038/s41586-023-06139-9
2. Horn L, Mansfield AS, Szcęsna A, et al. First-line atezolizumab plus chemotherapy in extensive-stage small-cell lung cancer. *N Engl J Med* 2018. doi:10.1056/NEJMoa1809064
3. Wherry EJ, Kurachi M. Molecular and cellular insights into T cell exhaustion. *Nat Rev Immunol* 2015. doi:10.1038/nri3862
4. Rudin CM, Poirier JT, Byers LA, et al. Molecular subtypes of small cell lung cancer: a synthesis of human and mouse model data. *Nat Rev Cancer* 2019. doi:10.1038/s41568-019-0133-9
5. Gay CM, Stewart CA, Park EM, et al. Patterns of transcription factor programs and immune pathway activation define four major subtypes of SCLC with distinct therapeutic vulnerabilities. *Cancer Cell* 2021. doi:10.1016/j.ccell.2020.12.014
6. Young MD, Behjati S. SoupX removes ambient RNA contamination from droplet-based single-cell RNA sequencing data. *GigaScience* 2020. doi:10.1093/gigascience/giaa151
7. Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J R Stat Soc B* 1995. doi:10.1111/j.2517-6161.1995.tb02031.x
8. Mann HB, Whitney DR. On a test of whether one of two random variables is stochastically larger than the other. *Ann Math Stat* 1947. doi:10.1214/aoms/1177730491
9. Szklarczyk D, Kirsch R, Koutrouli M, et al. The STRING database in 2023: protein-protein association networks and functional enrichment analyses for any sequenced genome of interest. *Nucleic Acids Res* 2023. doi:10.1093/nar/gkac1000

Package sources for the added sections:

`data/sources/poster_final.pdf` (our 24JSDP poster, P25; geneformer-lung-tcell poster/poster_final.pdf, untracked; sha256 94638dba...)

`data/sources/JSDP_P25_talk.pdf` (the talk that accompanied the poster; geneformer-lung-tcell talk/JSDP_P25_talk.pdf, untracked; sha256 3f013a5d...)

`data/git/dd7366c/scll_validation/perturbation_workflow/METHODS.md` (geneformer-lung-tcell @ dd7366c)

`data/git/53adee2/.../results_s100_luad_20260928/RESULT.md` (geneformer-lung-tcell @ 53adee2; merged to main in PR #37, 4455914)

`data/git/53adee2/.../results_s100_luad_20260928/primary_results.json` (@ 53adee2)

`data/git/53adee2/.../s100_isp/analyse_s100_luad_20260928.py` (@ 53adee2): gene groups, Ensembl IDs, ambient-risk values

the lab's ISP outcome standard (ISP-STD-1 v1.1, the floor's wording rules; not in this package)

Per-slide sources, from `docs/lung_tcell_talk_imrad_sources_20260925.md` (commits are in the geneformer-lung-tcell repository).

Slide	Section	Claim	Source
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1	Title	—	—
2	Background	(framing only)	—
3	Intro I1	4 original hits, tissue validation	talk/JSDP_P25_talk.pptx/pdf
4	Intro I2	Poster/talk hedges, quoted	poster/poster_final.html; talk/JSDP_P25_talk.pdf p.7
5	Methods M1	91.9%/F1 0.903 + 1-donor caveat; 42-vs-45	README.md; perturbation_workflow/METHODS.md lines 57– 67, 111–112, 33–39
6	Methods M2	ISP mechanics, six comparisons	README.md
7	Methods M3	Concordance criterion, verbatim	primary_test_perturbation/scripts/ build_delete_overexpress_shift_report.py @ b99365b
8	Methods M4	Ambient/donor-weighted methodology	ambient_risk_diagnostic.py line 198 @ b99365b; immune_axis_test/donor_robustness.py @ 9ec518c
9	Results R1	4 hits + spatial validation; cell-set caveat on the concordance half (2026-09-25)	talk/JSDP_P25_talk.pptx/pdf; targeted_panel_delete_overexpress_merged .csv @ 6882627 (overexpress_n 2,424 constant; delete_n 202/438/282/1,131); runner fix 66d235b (2026-09-22); no later commit to the results table
10	Results R2	JSDP's own QC + pending-test flag	talk/JSDP_P25_talk.pdf, slide 7
11	Results R3	p=0.216 Monte Carlo (B=100,000); PleuralEffusion/HTA8_2001; 2/35 exact-enumeration floor	t6_weighting_reconciliation.csv, t6_permutation_tests.csv @ 9ec518c; METHODS.md lines 29–32
12	Results R4	S100A8/S100A9 FDR + concordance failure	isp_plausibility_top_candidates.csv @ b99365b
13	Results R5	Ambient audit, 60/120 union	same CSV + ambient_risk_manifest.json @ b99365b
14	Results R6	NBEAL1	same CSV @ b99365b
15	Results R7	bf16 + frozen-FAIL gate	committed_run_stats/* @ origin/main; RESULTS_BF16.md @ afa0564
16	Results R8 (designated cut)	STRING mechanism	talk/JSDP_P25_talk.pdf, slide 10
17	Discussion D1	Synthesis	(rolls up the above)
18	Discussion D2/D3	Limitations, closing	perturbation_workflow/METHODS.md lines 24– 25; talk/JSDP_P25_talk.pdf, slide 11

Data files shipped with this report's build package (data/MANIFEST.tsv).

File	Origin	sha256 (first 16)
git/b99365b/scic_validation/ primary_test_perturbation/ tables/ isp_plausibility_top_candidates .csv	scic_validation/primary_test_perturbation/tables/ isp_plausibility_top_candidates.csv (git b99365bd0a1d3fcf128256b962fab440033e739d (geneformer-lung-tcell))	5f271197973 ec01a
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